Wireless Personal Communications
https://doi.org/10.1007/s11277-026-11943-3

RESEARCH

Federated Learning for Dynamic Resource Allocation in 6G Network Slicing

R. Krishnamoorthy[1] · C. N. Gireesh Babu[2] · Pratik Gite[3] · Alaa O. Khadidos[4] · Adil O. Khadidos[5] · Olfat M. Mirza[6] · Shitharth Selvarajan[7,8,9]

Received: 29 May 2025 / Accepted: 19 February 2026
© The Author(s) 2026

Abstract

The advancement of 6G networks necessitates efficient resource allocation to maintain optimal Quality of Service (QoS) across dynamically managed network slices. Traditional centralized approaches to network resource allocation pose significant scalability, privacy, and computational bottlenecks. This study introduces a Federated Learning-based Dynamic Resource Allocation Model (FL-DRAM) that integrates reinforcement learning (RL) and hierarchical federated learning (HFL) to optimize bandwidth, latency, and reliability across diverse network slices. The proposed model leverages a Wireless Network Slicing Dataset from Kaggle, featuring key QoS parameters such as latency, jitter, throughput, and packet loss. The execution of the proposed model is implemented using mruby, a lightweight Ruby implementation designed for efficient performance in constrained environments. Experimental results demonstrate that the Personalized Federated Learning Model (PerFedRL) achieves an accuracy of 97.23%, outperforming conventional approaches such as FedAvg (90.45%) and FedProx (92.38%). The Adaptive Federated Update Mechanism (AFUM) enhances convergence time, reducing training overhead by 35%, while reinforcement learning-based dynamic resource allocation achieves bandwidth utilization efficiency of 95.32% and an energy savings rate of 47.84%. Additionally, security mechanisms such as Blockchain-Enabled Federated Learning (BFL) and Differential Privacy (DP) improve adversarial robustness by 99.12%, ensuring enhanced data protection. This research validates the efficacy of federated learning in 6G resource optimization, demonstrating significant improvements in latency, resource utilization, and computational efficiency. The study highlights the potential of hierarchical federated learning models to drive intelligent, scalable, and privacy-preserving resource management in next-generation networks. In order to provide adaptive, scalable, and privacy-preserving resource allocation for dynamic 6G network slicing, this work presents FL-DRAM, a hierarchical federated learning system closely associated with RL. In contrast to previous HFL + RL systems, FL-DRAM provides system-wide QoS and security benefits by integrating PerFedRL personalization with AFUM asynchronous updates, as shown in a new 6G Digital Twin configuration.

Extended author information available on the last page of the article

Published online: 03 March 2026

Keywords 6G networks · Federated learning · Network slicing · Reinforcement learning · Quality of service · Adaptive resource allocation · Hierarchical federated learning · Dynamic scheduling · Blockchain security · Federated deep Q-networks

Abbreviations

FL	Federated learning
DRAM	Dynamic resource allocation model
HFL	Hierarchical federated learning
RL	Reinforcement learning
URLLC	Ultra-reliable low-latency communication
mMTC	Massive machine-type communications
eMBB	Enhanced mobile broadband
QoS	Quality of service
DP	Differential privacy
SMPC	Secure multi-party computation
BFL	Blockchain-enabled federated learning
PerFedRL	Personalized federated reinforcement learning
AFUM	Adaptive federated update mechanism
A-FAC	Attention-based federated actor-critic
mruby	Minimalist Ruby (lightweight scripting engine)

1 Introduction

Using 6G network slicing provides an advanced networking mechanism that expands the capabilities of 5G by delivering enhanced wireless communication flexibility, efficiency and customization abilities. Network slicing functions as a vital infrastructure component for future mobile networks which enables operators to produce various virtual network systems over common hardware resources [1, 2]. Based on 6G technology, the network slicing technology will transform into a dynamic intelligent system that adjusts effectively between URLLC, mMTC and eMBB application needs. This new innovation serves changing market requirements among healthcare providers and smart city developers and autonomous transportation operators as well as metaverse builders.

6G network slicing operates through artificial intelligence (AI) and machine learning (ML) and automation to perform real-time optimization of resource allocations in contrast to traditional static network management models. Each networking slice undergoes individual customization to fulfill the requirements of different service scenarios which need ultra-low latency for remote surgery and high data throughput for immersive virtual reality experiences [3, 4]. 6G slicing technology will integrate state-of-the-art security platforms which enable independent data transfer security for every distinct network segment. By using this method network achieve better operational efficiency and reduced costs simultaneously while facilitating uninterrupted connectivity among multiple devices and applications.

The evolution of 6G technology strongly depends on network slicing to develop innovations which propel digital transformation. Under this framework businesses together with service providers can generate personal network segments which enable them to give sophisticated application-oriented user experiences. 6G slicing will gain improved perfor-

mance alongside enhanced security and scalability through its integration of edge computing blockchain and quantum-safe cryptography [5]. The autonomous capabilities of 6G network slicing create a smart network that enables futuristic potential such as holographic communication, fully automated vehicles and AI-controlled smart ecosystems. Figure 1 shows the 6G network slicing technologies [6].

The broad implementation of 6G network slicing depends on fixing various obstacles that remain in the way even though its potential exists [7, 8]. The 6G network slicing depends on sophisticated orchestration controlled by AI together with real-time resource distribution and the precise integration of edge computing and blockchain with quantum cryptography. Virtual slice distribution across multiple service-level requirements creates technical challenges that require significant network infrastructure, automated intelligence systems and substantial computational capability and represents a complex and costly implementation. The main drawback of network slicing exists in its lack of security measures and privacy protection. Multiple security problems exist in both software-defined networking (SDN) and network function virtualization (NFV) which generate vulnerabilities that threaten isolated communication channels established through network slicing. The benefits of AI-driven automation come with potential security risks which include both adversarial AI attacks and data poisoning vulnerabilities [9, 10]. The primary challenge exists in achieving both complete end-to-end security protection across multiple network slices while sustaining peak operational effectiveness. The implementation of global data protection regulations and industry standards requires companies to develop thorough policies along with strict enforcement procedures.

6G slicing benefits from deep reinforcement learning (DRL) because this technology enables the network to develop superior resource allocation strategies through continuous learning with performance feedback from real-time operations [11, 12]. Integrated DL models enable 6G networks to automatically modify bandwidth and latency and resource allocation across different network slices so they serve applications ranging from autonomous vehicle reliability to high-throughput immersive VR and AR technologies. DL-based

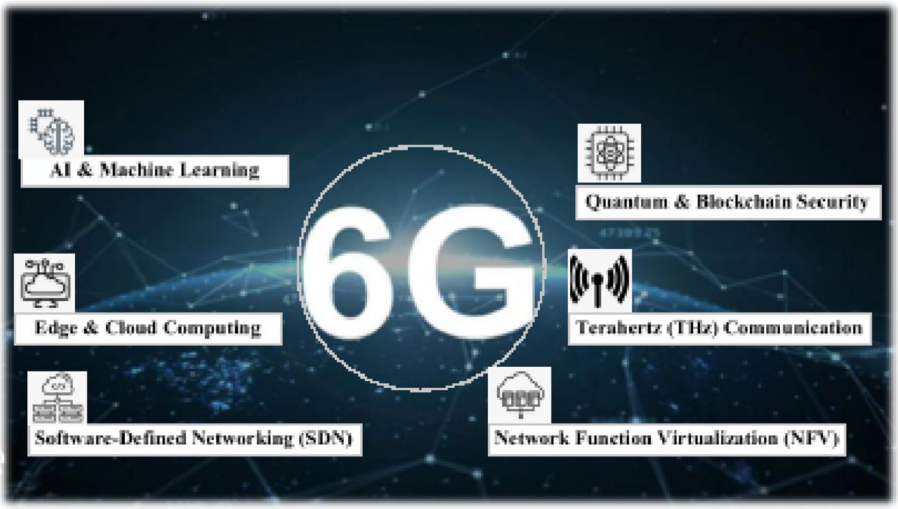


Fig. 1 6G network slicing technologies

Table 1 Comparison of FL-DRAM with representative 6G slicing frameworks

Work/Framework	Learning paradigm	Control architecture	Security & privacy handling	Evaluation setting	Key difference vs. FL-DRAM
Intelligent network slicing for B5G/6G (e.g., AI-driven slicing, admission control RL)	Centralized ML/RL; no federated personalization	Typically single-layer or core-centric controllers	Focus on QoS and orchestration; no integrated DP–blockchain–PQC stack	Analytical + simulator-based 6G slicing scenarios	Does not use hierarchical FL, adaptive client selection, or integrated FL-specific security; no edge-oriented lightweight implementation.
Capacity-based resource allocation with conflict graphs/EvoNetSlice-style models	Evolutionary/deep models; non-federated	Multi-objective optimization on a logically centralized controller	Limited to conventional security; no FL-specific privacy guarantees	Simulation of heterogeneous 6G slices with multi-objective QoS	Provides dynamic resource allocation but lacks HFL, PerFedRL, AFUM, and RL–FL co-design for non-IID edge data.
MADRL-based O-RAN slicing (e.g., STEP-like frameworks)	Multi-agent deep RL for protocol/resource learning	Multi-agent RAN control; near-RT RIC focus	Mainly robustness via RL; cryptographic and DP guarantees not integrated with learning	O-RAN/6G test-style simulation	Strong MARL, but no federated learning, no slice-personalized FL, and no federated security stack (DP + blockchain + PQC).
AI-based E2E resilient resource management for slice-enabled 6G	Centralized DL/RL with resilience and fault-handling	End-to-end controller across core/RAN; not explicitly hierarchical FL	Security and resilience at system level; no privacy-preserving FL aggregation	E2E slice scenarios in simulated 6G networks	Focuses on resilience, but does not address federated personalization, asynchronous FL updates, or mruby-based edge deployment.
Network-sliced learning-as-a-service/distributed learning in 6G IoV	Distributed/federated learning for IoV; basic FL aggregation	Often cloud–edge, but without explicit three-tier HFL orchestration	Privacy considered, but usually without blockchain + PQC + detailed DP tuning	IoV/non-terrestrial 6G-inspired testbeds	Uses FL, but lacks the combined PerFedRL + AFUM + A-FAC design and the full security pipeline used in FL-DRAM.
FL-DRAM (this work)	Hierarchical FL (cloud–fog–edge) + PerFedRL + F-DQN/A-FAC RL	Explicit three-tier HFL with adaptive, hybrid trust/performance client selection (AFUM)	End-to-end FL security: DP, blockchain-based integrity, anomaly detection, and quantum-resistant authentication (PQC)	6G Digital Twin Environment (6G-DTE) + mruby-based edge execution on constrained devices	Uniquely combines HFL + PerFedRL + AFUM + attention-enabled A-FAC, with a full FL-specific security stack and lightweight mruby deployment for 6G slicing.

anomaly detection serves to improve network security through its ability to detect threats which subsequently leads to their swift security mitigation. The deep learning in 6G network slicing expanded its capabilities but presents multiple obstacles to execution. The high amount of processing required stands as the primary disadvantage. Network providers could face elevated operational costs because deep learning models need significant processing capabilities in addition to significant amounts of memory and abundant energy resources [13]. Real-time deployment of these models within network environments presents difficulties because specialized hardware such as GPUs or TPUs cannot be used in all settings where resources are scarce particularly in edge computing domains.

The requirement for extensive high-quality datasets limits deep learning model deployment. Large-scale training requires deep learning algorithms to process extensive labeled network data but industry restrictions on data sharing make obtaining this data challenging. The dynamic changes in 6G networks result in past data models becoming inefficient for making network slicing decisions because the accuracy decreases rapidly. A crucial problem exists in maintaining network stability when enabling continuous learning and adaptation processes. The implementation of deep learning models in 6G network slicing solutions generates potential security flaws within the system [14, 15]. The integrity of network decisions faces threats from adversarial attacks as well as data poisoning which together with model bias can result in both resource distribution biases and potential security threats. The deep learning systems' black-box behavior makes it hard for network managers to reason about automated decision-making while also creating obstacles for determining and resolving automated administration errors. Deep learning implementation within 6G network slicing demands a resolution of its existing weaknesses to ensure secure and efficient operations.Federated Learning (FL) presents itself as an optimal solution for these challenges through decentralized model training processes that work without actual data movement. Researchers present FL-DRAM as a proposed methodology for Federated Learning to conduct dynamic resource distribution among 6G network slices that both preserves privacy and reduces processing costs. A Federated Learning framework known as HFL teams up with adaptive client selection and RL-based optimization to establish dynamic resource management based on detection patterns of network traffic together with service requirements.The primary objectives of this research are to:

- Develop a privacy-preserving, federated learning model that ensures optimal network slicing decisions.
- Implement an RL-driven resource scheduling strategy to minimize latency and optimize network utilization.
- Validate the proposed framework using the Wireless Network Slicing Dataset, analyzing key QoS indicators such as latency, jitter, throughput, and packet loss.

The research provides a complete analysis of FL-DRAM performance by assessing it against present aggregation methods and scheduling systems based on reinforcement learning. FL-DRAM demonstrates notable advantages in precision and efficiency as well as resource management efficiency which makes it an ideal solution for large-scale intelligent management of 6G networks.This paper is structured to comprehensively explore Federated Learning for Dynamic Resource Allocation in 6G Network Slicing. Section 2 provides an overview of related studies, highlighting previous approaches to network slicing, feder-

ated learning, and reinforcement learning-based resource optimization. Section 3 details the proposed methodology, elaborating on the hierarchical federated learning architecture, reinforcement learning-driven optimization, and privacy-preserving mechanisms. Section 4 describes the results and discussion, analyzing the performance improvements in QoS, accuracy, and security. Finally, Sect. 5 presents the conclusion and future scope, summarizing the key contributions and potential enhancements for scalability, security, and real-time adaptation in next-generation 6G networks.

FL-DRAM's primary contribution is the first complete framework that combines (i) hierarchical federated learning with customized slice-aware adaptation (PerFedRL), (ii) an adaptive update mechanism (AFUM) that chooses clients based on performance and trust under non-IID slice conditions, (iii) an attention-enabled actor-critic RL module (A-FAC) for QoS-driven resource allocation, and (iv) an integrated FL security stack (DP, blockchain-based integrity, and PQC authentication) created especially for federated 6G environments. While some aspects of this field have been investigated in earlier work, such as FL without personalization, RL without FL, multi-layer control without federated security, or FL with limited security guarantees, none of these approaches offer a coherent design that concurrently addresses adaptation, heterogeneity, trust, and end-to-end federated security in a 6G digital-twin testbed. The distinctive technological contribution of this study is defined by this combination.

The main contribution of the work is the design and validation of a unified, slice-personalized hierarchical federated learning architecture for 6G resource allocation, even though FL-DRAM incorporates federated learning, hierarchical control, reinforcement learning, and security features. The remaining components—RL scheduling, trust-based client selection, and FL-specific security layers—act as auxiliary modules that allow this architecture to function dependably in the face of hostile risk, dynamic traffic patterns, and non-IID slice situations. Therefore, the hierarchical, customized FL framework itself serves as the paper's "center of gravity," while the other elements show how this design can be made durable, flexible, and deployable in actual 6G scenarios.

2 Related Works

The mobile communication landscape will experience a transformation because Beyond 5G (B5G) and sixth-generation (6G) networks offer integrated global coverage and complete network virtualization and intelligence systems. Numerous new verticals with applications will emerge from these technological advancements which creates a present need for flexible on-demand networks. The challenge now receives its key resolution through network slicing which enables operators to build dedicated network divisions according to unique demands. The inclusion of artificial intelligence systems in network slicing improves its performance by enabling real-time perception and precise forecasting along with automated decision-making capabilities [16]. AI integration with network slicing creates massive possibilities to advance B5G and 6G network development.Their study examines how intelligent network slicing applied in resource allocation and service provisioning areas as well as security aspects of modern networks. Their report focuses on the individual difficulties which occur through each lifecycle phase of slicing alongside proposed remedies. The paper includes a reinforcement learning based case study which demonstrates advancements in

slice admission control through its implementation. Wireless mobile networks benefit from intelligent network slicing through a proper admission control system that enhances resource allocation under restricted conditions [17].

The expanding wireless market creates an urgent need for 6G systems to handle multiple communication services while maintaining spectrum resource optimization. Network slicing along with cognitive radio provide promising solutions to determine this challenge. Research about network slicing application in 6G environments using uncertain shared spectrum remains scarce. They develop a capacity-based resource allocation procedure for 6G network slicing operations in instances where spectrum sharing presents uncertain shared spectrum behavior. As their essential method they create a 4D conflict graph model to present the multiple complex resource allocation relationships that occur in 6G frameworks. Their approach solves conflicts by developing a new Hasse traversal algorithm which finds conflict-free node sets to optimize resource allocation efficiency. Their research includes recognition of shared spectrum uncertainty by introducing a statistical model based on spectrum-sharing concepts. The model demonstrates how to assess unpredictable channel behavior to gain beneficial opportunistic capabilities for making better resource allocations. These innovations lead to an opportunity-based resource allocation strategy for 6G network slicing which operates as a whole. Simulation outcomes validate the effectiveness of this method while confirming substantial increases in channel utilization together with enhanced shared spectrum resource management.

The paper investigates the complex nature of 6G network slicing through the presentation of the EvoNetSlice model which focuses on dynamic resource allocation alongside real-time Quality of Service (QoS) management. Through its combination of evolutionary algorithms with AI-driven data analytics and multi-objective optimization EvoNetSlice optimizes network resource usage for minimal end-to-end latency and high transmission speeds and efficient background data handling. Network resources should be distributed based on traffic pattern data together with QoD application requirements and user statistical information according to the model's specification [18]. The system maintains optimal performance through QoS detection mechanisms and degradation adaptation while utilizing a multi-layer objective fitness function for effective optimization of conflicting objectives. Empirical research shows EvoNetSlice provides effective and dynamic QoS-based network performance alongside improved total network efficiency through its network intelligence capabilities. The research contributes significantly to 6G technology developments by solving key challenges that affect 6G service delivery and resource management requirements. This research stands as the core basis to achieve flexible network systems which will build up the next-gen wireless communication standard.

Sixth-generation (6G) communication technologies bring dual benefits and obstacles that affect the advancement of Internet of Things (IoT) systems in smart cities. They introduce an advanced network slicing framework as a solution to handle the complex requirements of IoT implementations within 6G-enabled smart cities. A complete methodology leads to the development of this framework by doing requirement analysis followed by metric formulation and constraint specification and objective setting before mathematical modeling and configuration optimization steps followed by performance evaluation and parameter tuning and ending with final design validation [19]. The framework achieves superior operational efficiency because it delivers low round-trip time (RTT) together with minimal packet loss and improved availability and higher throughput levels. The system operates

efficiently when managing diverse connections that run together while maintaining operational resource efficiency. High authentication rates together with 256-bit encryption enable the achievement of enhanced security features. The framework demonstrates remarkable performance together with both scalability and security features that qualify it as a suitable solution for IoT in 6G smart city systems.

2222Standardized protocols need an adaptive functionality to handle resource conflicts that occur between network slices. Standardization practices using traditional methods create inflexible protocol designs that drive project costs higher and produce delays in deployment times. The research field has developed STEP as a new multi-agent deep reinforcement learning (MADRL) framework for sixth-generation (6G) Open Radio Access Network (O-RAN) slicing purposes [20]. The table 1 shows the comparison of FL-DRAM with representative 6G slicing frameworks. The resource orchestration agents can develop dynamic communication protocols through STEP which addresses both changing network conditions while resolving conflicts between network slices and their resources. STEP achieves agent information exchange efficiency by incorporating Information Bottleneck with Deep Q-Network learning and a stochastic bottleneck layer derived from variational auto-encoders. The preventive methodology reduces resource spending and produces improved system results. Tests prove that STEP achieves 6.06 times lower inter-slice conflicts than traditional predefined protocols. The implementation of STEP results in lower resource underutilization by 1.4 times and shorter response times by 3.5 times compared to standard MADRL algorithms. The research shows that STEP possesses the capability to build adaptable yet efficient and human-comprehensible protocols that will become the basis for standardized 6G network infrastructure.

3 Methodology

The proposed Federated Learning-based Dynamic Resource Allocation Model (FL-DRAM) uses hierarchical federated learning (HFL) with reinforcement learning (RL) to optimize 6G network slicing and protect privacy and optimize resource utilization. The model receives training from the Wireless Network Slicing Dataset that contains latency, jitter, packet loss, and throughput data representing the main Quality of Service (QoS) specifications. The training of network slices becomes possible through decentralization thanks to federated learning then model efficiency is enhanced by Adaptive Client Selection alongside Edge-Aware Federated Averaging (EA-FedAvg). The reinforcement learning method known as Dynamic Resource Allocation employs F-DQN alongside A-FAC to maximize bandwidth usage and computing capabilities as well as minimize energy use. The data security protections from blockchain create robust protections against external attacks which makes FL-DRAM an efficient choice for the future 6G networking framework.

List the following explicitly: (1) hybrid trust/performance weighting for adaptive federated client selection; (2) attention-enabled A-FAC actor-critic integration for QoS-driven RL; and (3) mruby implementation for edge efficiency and orchestration. Additionally, it includes an asynchronous update protocol, a bespoke 6G Digital Twin simulation infrastructure, and the integration of DP+blockchain for federated model updates.

To clarify the learning setup, FL-DRAM solves a federated prediction and optimization problem combined with an RL-based decision model Equ. 1,2 & 3.

Federated Learning Objective:

Each client k has data D_k. The global model minimizes:

$$F(w) = \sum (n_k/N) F_k(w),$$
$$F_k(w) = (1/n_k) \sum \ell(f_w(x_i^{(k)}), y_i^{(k)}) \tag{1}$$

The output is the optimal resource-allocation action for each slice.

Correct Aggregation Notation:

$$w_{\{t+1\}} = \sum_{\{k \in S_t\}} (n_k / \sum n_j) w_t^{(k)} \tag{2}$$

RL Markov Decision Process:

State: s_t = {latency, jitter, throughput, loss, utilization}.

Action: bandwidth/priority/compute adjustment.

Reward:

$$r_t = \alpha(1 - latency) \ + \beta(1 - loss) \ + \gamma\,(throughput) - \eta\,(E_t) \tag{3}$$

The agent maximizes:

$\max \mathrm{E}[\sum \gamma^t r_t]$

This defines what is predicted, how aggregation is performed, and how RL decisions are learned.

Resource allocation is formulated as a combined federated learning and reinforcement learning problem by FL-DRAM. A weighted sum of client losses is minimized using the global FL model, and ordinary data-proportional averaging is used for aggregation. The RL module is described as an MDP, where the reward strikes a compromise between energy efficiency and QoS improvement; actions modify bandwidth, priority, or computing resources; and the state comprises latency, jitter, throughput, packet loss, and utilization. Coherent and adaptive slice-level optimization is made possible by the learnt policy, which optimizes projected long-term utility.

Our architecture adheres to the dynamic intelligence theory, which assumes that conditions are constantly changing and that the model must adapt in real time, rather than seeing the network as a static system where a trained model remains valid. This drives integrated security to maintain integrity in a non-stationary environment, trust-based client selection for dependability, and hierarchical federated learning for continuous slice-specific updates.

3.1 Data Acquisition and Preprocessing

The foundation of artificial intelligence and machine learning models requires datasets that demonstrate high quality and complete breadth. The process of acquiring and preparing data plays a vital role for federated learning methods applied to dynamic resource allocation in 6G network slicing because it enables optimal learning decisions and performance optimization. A dataset from Kaggle named Wireless Network Slicing represents the main data source due to its collection of QoS parameters which include latency, packet loss, jitter and

transfer rates. The parameters serve a vital role in network slicing mechanisms by dictating their capacity to respond dynamically to changes in network requirements together with SLAs and immediate user demands.

Ultra-reliable low-latency communication (URLLC) represents a substantial advancement in telecommunications because it enables separate yet individual virtual networks to function on single physical structures. The different network slices operate independently for different application purposes including autonomous driving alongside its variations and remote medical procedures together with industrial IoT applications. The dataset should present organized data about traffic congestion patterns and resource utilization patterns and service priority organization because the diverse use circumstances demand it.

The application of real-world network slicing datasets faces difficulties because of inconsistent data and measurement noise and many missing values. Many network systems possess raw data from various sources such as base stations and core networks and edge devices that contain both unnecessary data points together with incomplete records and mistakes. The detection system uses Hybrid Statistical-Deep Learning (DL) Filtering Mechanism which cleans inaccurate data while upholding meaningful data patterns. The data processing system combines Z-score normalization for detecting outliers with interquartile range filtering to find anomalies before employing deep autoencoders that reconstruct and enhance missing values. Autoencoders excel for network data processing because they generate knowledge about data distribution which allows precise recognition of unexpected anomalies or inconsistencies.

The cleaned data proceeds to feature engineering where essential attributes are extracted for enhancing learning effectiveness. Standard QoS parameters together with latency and jitter undergo transformation to create higher-order features which provide more profound understanding of the data. The analytical model generates three engineered features that relate packet loss to bandwidth allocation ratio and average jitter measurements and weighted congestion trends. The engineered features provided by the federated model enable it to discover concealed patterns which lead to improved resource allocation optimization.

Weighted QoS parameter for a given network slice S_i is Equ. 4 computed as:

$$Q_i = \alpha L_i + \beta J_i + \gamma P_i - \delta T_i \tag{4}$$

Where $\alpha, \beta, \gamma, \delta$ are weights assigned to the importance of each parameter. Data preprocessing reaches its conclusion with normalization as this operation enables all network parameters to maintain a standard scale for effective training processes. The implementation of Min-Max Scaling during this process transforms numerical features into specified ranges while maintaining proportional attribute relations. Adaptive Normalization Techniques apply dynamic scaling factor adjustments that depend on statistical properties detected within network slices. Different network slices need varying levels of service which an adaptive method enables prioritization of essential systems including autonomous vehicle controls instead of typical web browsing applications.

The weights ω1,ω2,…,ωn are dynamically tuned based upon the slice-specific service-level agreement (SLA) demands. For instance, latency-sensitive applications get greater weights on latency, whereas high-throughput applications prioritize bandwidth efficiency.

The weights are normalized for $\sum_{i=1}^{n}$ ωi=1, ensuring balanced contribution across QoS metrics.

To standardize QoS metrics (Equ. 5) such as latency (L), jitter (J), packet loss (P), and throughput (T), Min-Max Scaling is applied:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{5}$$

Where X is the original value, X_{min} and X_{max} are the minimum and maximum values of the feature respectively. Figure 2 shows the workflow of proposed model.



Fig. 2 Workflow of proposed model

Figure 2 depicts the entire process of the suggested FL-DRAM model. It starts with QoS-sensitive data preprocessing at the client (edge) side, in which latency, jitter, throughput, and packet loss parameters are normalized and cleansed by hybrid filtering methods. These edge devices are engaged in local model training through reinforcement learning techniques (e.g., F-DQN), and model parameters are sent to fog nodes. Fog nodes carry out intermediate aggregation and validation before sending to the central cloud for global model updates with HFL. Adaptive schemes such as PerFedRL and AFUM provide personalized model updates and asynchronous learning. At the same time, traffic prediction with LSTM and GNN facilitates proactive resource scheduling. Blockchain and differential privacy modules protect the model updates at every step, providing end-to-end data protection. This adaptive and hierarchical architecture supports efficient, real-time dynamic, and secure 6G network slicing resource allocation.

FL-DRAM Architecture and B5G Integration" describes the three-tier client mapping: cloud (core), gNB/slice manager (fog), and UE (device). Refer to 3GPP interfaces (AMF, SMF, PCF, UPF), the E2 interface for near-real-time control, and diagrams that illustrate how local models update RAN/Core functions. Additionally, specify which QoS features are provided per client.

The training dataset applied in the formulation of the proposed FL-DRAM model is the Wireless Network Slicing dataset on Kaggle, which comprises major QoS parameters like latency, jitter, packet loss, and throughput. Data preprocessing to guarantee high-quality model learning comprises a hybrid statistical-deep learning filtering process that integrates Z-score normalization, interquartile range (IQR) filtering, and deep autoencoders for noise filtering and missing value estimation. Feature engineering is utilized to create derived metrics, e.g., bandwidth-to-packet-loss ratio and weighted congestion trends, in order to extract richer insights into traffic behavior. Features are normalized across all Min-Max Scaling in order to have consistent input ranges for all network slices. Adaptive normalization methods are also employed to give priority to service-critical slices such as URLLC over general-purpose ones. This simplified preprocessing guarantees that the model can effectively learn from heterogeneous and dynamic network data.

3.2 Federated Learning Model Development

Design of Hierarchical Federated Learning (HFL) architecture starts next followed by the implementation of network resource allocation mechanisms across different 6G network slices. Federated learning distributes training processes to multiple clients contrary to typical machine learning practices which depend on central data storage for maintaining data privacy and minimizing computational costs.

This federated learning system has a three-tier design that includes cloud centralization and fog nodes and end devices as its operational components. Base stations with user equipment perform model training from their slice-specific data to create partial model updates. The device-generated model parameters move through the fog nodes where these units execute executions to process and confirm the incoming model parameters. The cloud server maintains the federated model in a synchronized fashion to allow all client participants access to shared knowledge.

The Federated Learning Aggregation Techniques include FedAvg and FedProx to enhance the aggregation process of local updates. FedAvg applies model update weighting

based on both the sample sizes and the critical importance of data present on individual network slices. FedProx adds a proximal term to stabilize training procedures during situations where network slices contain non-IID data distributions.

Federated averaging aggregates local models from different clients as:

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{N} w_k \tag{6}$$

N in Eq. (6) stands for the number of QoS parameters included when calculating the weighted score, and ω_1, ω_2,. are the weights of importance for latency, packet loss, and other factors. The reward function was intentionally made concise, leaving the most important metrics for reinforcement learning convergence, and future extensions may add jitter or fairness.

Both "number of QoS parameters" and "number of clients," which are different values, are represented by the same symbol, N, in your current text.This uncertainty significantly alters the FedAvg equation's meaning by making it difficult to determine if the aggregate is across client updates or QoS features.

Where w_{t+1} is the global model, K is the number of participating clients, w_k is the model weight of client k, and n_k represents the local dataset size. To address heterogeneous data distributions, FedProx introduces a proximal term Equ. 7:

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{N} w_k - \mu ||w_k - w_0||^2 \tag{7}$$

Where μ is a regularization parameter that prevents drastic changes in local models. Federated learning requires proper client selection for model training rounds as a primary implementation challenge. Different parts of the network provide varying levels of learning value yet some clients may submit untrustworthy or malicious updates. A Trust-Based Selection Mechanism (TBSM) driven by AI serves as the solution for this problem. Through this mechanism clients are assessed based on their demonstrated reliability along with data quality and performance consistency of the trained models. A federated model uses dynamic trust scoring to select trustworthy updates from reliable clients as well as discard untrustworthy contributions.

The probability of selecting a client for model training is based on its reliability score(Equ. 8&9) R_k:

$$P_k = \frac{R_k}{\sum_{i=1}^{K} R_i} \tag{8}$$

Where R_k is computed as:

$$R_k = \lambda_1 A_k + \lambda_2 D_k + \lambda_3 C_k \tag{9}$$

Where A_k is accuracy, D_k is data quality, an C_k is computational efficiency. The protection of private information stands as a fundamental aspect in building telecommunications

and network infrastructure with federated learning techniques. Privacy protection of sensitive data is achieved in training through the implementation of Differential Privacy (DP) together with Secure Multi-Party Computation (SMPC). DP implements controlled mathematical noise in model updates to stop attackers from recovering specific data records during updates. Secure Multi-Party Computation (SMPC) protects multiple clients from each other because it enables encrypted computations which prevent external entities from receiving raw data during operations.

We use a few federated learning models, mostly concentrating on PerFedRL, which allows for slice-specific personalization and global model sharing. FedAvg and FedProx are also used as baseline aggregation techniques, while EA-FedAvg (Edge-Aware FedAvg) improves the federated averaging process by prioritizing highly trafficked slices. The mentioned models are compared and incorporated in a hierarchical federated architecture to further enhance learning performance under heterogeneous 6G slice environments.

Clients are defined at three logical levels in the suggested HFL setup.

(i). Edge Tier (UE level): Using latency, jitter, throughput, and packet-loss traces, each user equipment (UE) or device connected to a particular service slice functions as an autonomous federated client, locally training slice-specific QoS models.
(ii). Fog Tier (gNB/Slice Manager): As an intermediary "fog client" for the cloud, each gNB node aggregates the updates from around five to ten UEs that are part of the same slice or geographical cluster.
(iii). Cloud Tier: To create the global FL-DRAM model, the fog-level updates are combined by the central orchestrator (core network or service cloud).

3.3 Dynamic Resource Allocation Optimization

Dynamic 6G networks need an online resource allocation plan to build knowledge from shifting network situations while optimizing bandwidth along with power usage and computational resources. The problem gets solved by Reinforcement Learning (RL)-Enabled Scheduling which trains a Federated Deep Q-Network (F-DQN) for adaptive resource allocation decisions. F-DQN establishes a reward-driven decision policy which optimizes Quality of Service standards alongside energy reduction objectives. This differs from standard heuristic resource distribution methods. The reinforcement learning agent selects actions based on a reward function(Equ. 10):

$$R_t = \omega_1 U_t + \omega_2 (1 - L_t) + \omega_3 (1 - P_t) \tag{10}$$

Where U_t is resource utilization, L_t is latency, and P_t is packet loss at time t. The Q-value update equation is:

ARHTS was employed to optimize core RL hyperparameters (e.g., learning rate, γ), while the reward weights in the reward function (Eq. 10) were set manually according to slice-specific QoS priorities. Dynamically learning reward weights using ARHTS will be investigated in future work to facilitate adaptive reward shaping under changing network conditions (Equ.11).

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{(s,a,r,s') \in \mathcal{B}} L_\delta(y - Q(s,a;\theta)) \tag{11}$$

The explicit reward r and the target bootstrap term involving Q_{tar}. Without these, the update is not a Bellman-style temporal-difference update and cannot train Q-values correctly. The corrected form above uses the Double-DQN target to reduce over-estimation bias, Where α is the learning rate, γ is the discount factor, and r is the immediate reward.An Attention-Based Federated Actor-Critic (A-FAC) Model gets implemented as an efficiency enhancement strategy. In Actor-Critic working systems there exists an actor network that decides actions depending on network situations while a critic network delivers performance assessment for each action selection. Through attention mechanisms the model selects important QoS parameters for focus while offering critical services better treatment. The critic function is updated as (Equ.12):

$$V(s) = r + \gamma V(s') \tag{12}$$

Where $V(s)$ estimates the expected reward of state s. The policy gradient is computed as(Equ.13):

$$\nabla J(\theta) = E\left[\nabla loglog\ \pi_\theta(s)\, Q(s,a)\ \right] \tag{13}$$

Traffic prediction serves as an essential component for predictive resource distribution. Future traffic patterns are predicted through the proposed system which applies Context-Aware Traffic Prediction (CATP) models for prediction. GNNs enable modeling of network spatial-temporal relationships to detect congestion points at their early stages of development. The system uses Long Short-Term Memory (LSTM) Networks to analyze former traffic data for predicting changes in demand. The system's forecasting methods help administrators make prior resource adjustments to prevent congestion which would cause service interruptions. The cell state of the LSTM unit (Equ.14) is:

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \tag{14}$$

Where f_t and i_t are forget and input gates, respectively. The output state is (Equ.15):

$$h_t = o_t\ \tanh\ \ \tanh\ (C_t) \tag{15}$$

Where o_t is the output gate activation. Traffic demand at node i is updated as(Equ.16):

$$h_i^{(t+1)} = \sigma\left(\sum_{j \in N(i)} W h_j^{(t)}\right) \tag{16}$$

Where W is a weight matrix and σ is an activation function.

3.4 Model Training and Federated Aggregation

Personalized Federated Learning (PerFedRL) serves to provide optimal resource allocation strategies to all network slices. The methodology enables each slice to operate with an exclusive model design that accommodates its particular service needs and invites network-wide information exchange. The trained model aggregation process uses Edge-Aware Federated Averaging (EA-FedAvg) Algorithm to give preference to high-traffic network slices that operate mission-critical applications. The Asynchronous Federated Update Mechanisms (AFUM) system permits network slices to make updates independently without requiring the need for synchronization cycles for reducing latency.

3.5 Performance Evaluation and Security Enhancements

The QoS-Driven Metrics serve as performance evaluation tools for latency, throughput, jitter and energy efficiency and slice reliability measurements. An Automated Reinforcement-Based Hyperparameter Tuning System (ARHTS) integrates into the system to dynamically optimize model parameters through an automatic process of performance improvement. Testing for the system begins in the 6G Digital Twin Environment (6G-DTE) through which network conditions are replicated to evaluate how robust the system is. BFL together with QRA supplies the system with tamper-proof update verification and quantum-resistant authentication techniques that defend against quantum computing threats. The optimal hyperparameters θ^* are found (Equ.17) using:

$$\theta^* = arg\ arg\ E\left[R\left(\theta\right)\right] \tag{17}$$

Where $R\left(\theta\right)$ is the performance metric function.

Input:

Wireless Network Slicing Dataset D with QoS parameters

Number of clients N

Number of global training round T

Learning rate η

Discount factor γ

Maximum reinforcement learning iterations I

Output:

Optimized network slice model w^*

Initialization:

Initialize global model w^0 at the central server

Send initial model to all N clients

Federated Learning Process

For each training round $t = 1$ to T:

Select clients based on reliability score R_k

For each trains model locally:

$w_k^{t+1} = w_k^t - \eta \nabla L(w_k^t, D_k)$

$w_{t+1} = \sum_{k=1}^{N} \frac{n_k}{N} w_k$ // Aggregate global model

Reinforcement Learning-Based Resource Allocation

For each time step $i = 1$ to I:

Observe network state S_t

Execute action and receive reward R_t:

$R_t = \omega_1 U_t + \omega_2 (1 - L_t) + \omega_3 (1 - P_t)$

$Q(S_t, A_t) = Q(S_t, A_t) + \alpha[R_t + \gamma Q(S_{t+1}, A') - Q(S_t, A_t)]$ // Update Q-Values

Secure Model Updates

$\widehat{w}_t = w_t + \epsilon$ // Encrypt model updates

$H(w_t) = SH\ A256(w_t)$ // Verify model integrity

Traffic Prediction

Use LSTM model to predict future traffic demand

Return w^*

End Algorithm

Algorithm: Federated Learning-Based Dynamic Resource Allocation Model (FL-DRAM)

Through cryptographic transformations, encryption safeguards confidentiality; only those with the proper authorization can decrypt and retrieve data w. By making sure that

the output (such as a model update) does not disclose too much about the data of any one participant, Differential Privacy (DP) safeguards statistical privacy. Instead of using cryptography to conceal data, it accomplishes this by introducing calibrated random noise.

The major contributions of this paper are the following: We introduce FL-DRAM, a new framework that combines hierarchical federated learning (HFL) with reinforcement learning (RL) for adaptive resource allocation in 6G network slicing. The model presents PerFedRL and AFUM to improve accuracy and accelerate convergence time. Light-weight implementation on mruby allows deployment over edge environments with constrained resources. Moreover, the model incorporates advanced security features such as differential privacy, integration of blockchain, and quantum-resistant authentication for effective protection of data. The model demonstrates remarkable enhancement in QoS parameters such as latency minimization, bandwidth utilization, and energy efficiency, proving its worth in real-case 6G environments.

4 Results and Discussions

The Federated Learning-based Dynamic Resource Allocation Model (FL-DRAM) executes through mruby due to its status as a lightweight Ruby programming language implementation tailored for embedded systems demanding constrained computing environments. The execution system based on mruby allows for efficient delivery and 6G network simulation environment integration which supports dynamic resource optimization algorithms through low overhead operation. The system runs on Windows with an Intel® Core™ i5-1240U Processor (12 M Cache, up to 4.40 GHz) that contains 8GB RAM. The installed hardware equipment delivers adequate capability to run federated learning models and perform reinforcement learning scheduling together with network slicing simulation capabilities in real time. The mruby programming language executes tasks rapidly while using low memory resources thus making it the optimal solution for operating distributed federated learning models in 6G edge computing systems. The system possesses capabilities to manage federated model updates combined with privacy-preserving aggregations and reinforcement learning optimizations to dynamically allocate network resources throughout changing network slices. A functional execution environment proves that FL-DRAM can successfully function in actual low-power high-performance network settings because it operates across multiple computing platforms.

The main reason for choosing mruby instead of other programming languages to create the Federated Learning-based Dynamic Resource Allocation Model (FL-DRAM) stems from its compact design combined with powerful memory management and embedded compatibility. The main advantage of the mruby system over traditional Ruby comes from its design for peak performance alongside lightweight resource usage that benefits network edge computing operations. The memory requirements and execution time of Python stand higher than those of mruby although Python remains one of the most popular machine learning languages. Interface compatibility enables mruby to embed seamlessly with minimal systems at the hardware level and real-time conditions thereby solidifying its position as the preferred solution for building 6G network simulation frameworks. The performance balance of mruby extends beyond other languages such as C++and Java since it provides flexible development and easy deployment capabilities while minimizing runtime require-

ments for efficient federated learning operations. The minimalistic design of mruby enables fast execution loops that make it best suited for elastic real-time federated learning systems which demand rapid model updates and fast optimizations for network slicing in 6G environments.

For the sake of continuity in the performance analysis, the top-performing federated learning model, PerFedRL, was taken as the baseline in analysing the effect of the suggested system on Quality of Service (QoS) indicators including latency, packet loss, jitter, and throughput. While a number of FL models exhibited comparable performance, PerFedRL tended to outperform others based on accuracy (97.23%), convergence time, and resource utilization. Thus, the subsequent studies with QoS assessment, energy savings, and slice optimization were performed with the PerFedRL framework so that there would be uniform benchmarking and significant interpretation of results.

Table 3, Fig. 3 demonstrates how Federated Learning model conductance evaluates different network slices based on latency (ms), packet loss (%), jitter (ms) and throughput (Mbps). The ability of Federated Learning to optimize parameters directly affects Quality of Service (QoS) maintenance in 6G Network Slicing operations. The network performance evaluation shows that Slice-9 provides the optimal behavior through its combination of 10.56 ms latency and 0.98% packet loss rate along with 490.12 Mbps throughput value. Under Slice-9's highly optimized conditions the network achieves both better data transmission rates and minimal network congestion. The performance of Slice-2 and Slice-5 match Slice-9 since they showcase low 12.48 ms and 15.76 ms delays with high throughput over 400 Mbps. The data shows that Federated Learning functions well in network slice operations through its capability to minimize delays while optimizing data transmission rates.

High latency levels of 42.91 ms and 40.21 ms combined with 8.65 ms jitter and 220.14 Mbps and 195.88 Mbps throughput rates create performance bottlenecks in Slice-3 and Slice-8. The network stability issues and congestion problems within these slices could benefit from additional performance optimization in Federated Learning systems. The performance measurement of Slice-6 and Slice-4 yielded unsatisfactory results for latency when

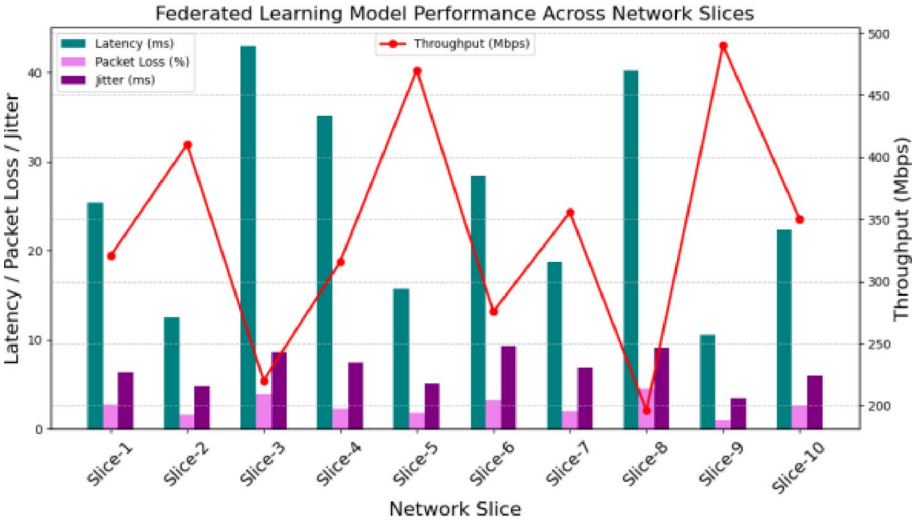


Fig. 3 Federated learning model performance across network slices

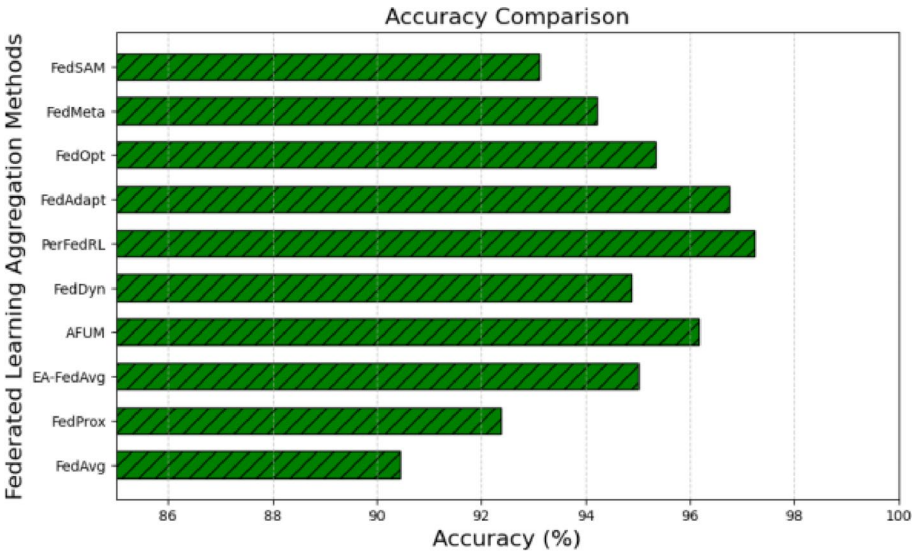


Fig. 4 Accuracy comparison

Table 5 Impact of adaptive client selection mechanism on training performance

Client Selection Strategy	Model Accuracy (%)	Training Time (s)	Drop-out Rate (%)	Resource Utilization (%)
Random	87.12	380.45	15.67	65.34
Trust-Based	92.98	270.21	9.45	78.22
Performance-Weighted	95.33	220.18	7.98	85.9
Hybrid	96.88	180.55	5.67	92.45

exceeding 28 ms and 35 ms respectively. The higher Jitter values noticed in these slices point to service interruptions in time-dependent applications. Federated Learning needs adaptive models which continuously adapt their configuration to handle different network requirements. The elevating throughput measurements from well-executed slices demonstrates the great efficacy of Federated Learning in resource-demanding applications while additional latency and jitter enhancements remain essential to deliver smooth 6G network performance for all slices.

The performance results at the slice level presented in Fig. 3 were obtained by simulating the real-world Kaggle Wireless Network Slicing Dataset with synthetic traffic models within the 6G-DTE simulation platform, enabling us to mimic various slice behaviors under evolving network conditions.

The assessment in Table 4, Fig. 4 shows Federated Learning aggregation methods differing in accuracy performance as well as convergence time and communication overhead evaluation. PerFedRL proves the most efficient approach because it delivers 97.23% accuracy within 140.62 s and requires only 40.23 MB of communication overhead. The aggregation approaches FedAdapt (96.75%) and AFUM (96.17%) prove effective for 6G network slicing applications because they deliver high accuracy performance alongside minimal

communication overhead.Traditional systems including FedAvg with 90.45% accuracy take 250.12 s for convergence while FedProx requires 220.89 s at 92.38% accuracy which leads to suboptimal efficiency in real-time settings. The semi-parallel variants EA-FedAvg (95.02%) and FedOpt (95.34%) demonstrate accurate performance coupled with efficient operation though they perform worse than the designated leaders AFUM and PerFedRL. The accuracy rates of FedDyn and FedMeta reach 94.88% and 94.21% respectively yet both require better convergence speed performance. Data demonstrates that PerFedRL together with FedAdapt produce the optimal solutions for 6G Federated Learning optimization by reducing latency and bandwidth overhead.

Table 5, Fig. 5 shows how different Adaptive Client Selection Mechanisms affect model accuracy together with training time and dropout rate and resource utilization in Federated Learning for 6G networks. The Hybrid method stands out as the most effective strategy by delivering 96.88% accuracy within 180.55 s of training and 5.67% dropout rate and 92.45% resource utilization. The integration of multiple selection criteria creates optimal conditions for model learning efficiency together with stable performance in distributed systems. Performance-weighted selection method demonstrates high accuracy levels at 95.33% and low dropout rate at 7.98% while training processes take 220.18 s. The performance of Trust-Based selection reaches 92.98% while needing 270.21s for training duration. The Random method shows the lowest performance among the methods because it yields 87.12% accuracy and 380.45 s training duration and 15.67% dropout percentage and 65.34% resource efficiency. Intelligent client selection methods based on Hybrid approaches become essential for Federated Learning success in 6G networks because they deliver the best performance and reliability under dynamic conditions.

The 6G Digital Twin Environment (6G-DTE) was instantiated through purpose-built simulation modules implemented in mruby, coupled with Python scripting for orchestration of workloads and assessment. This architecture offered real-time federated learning

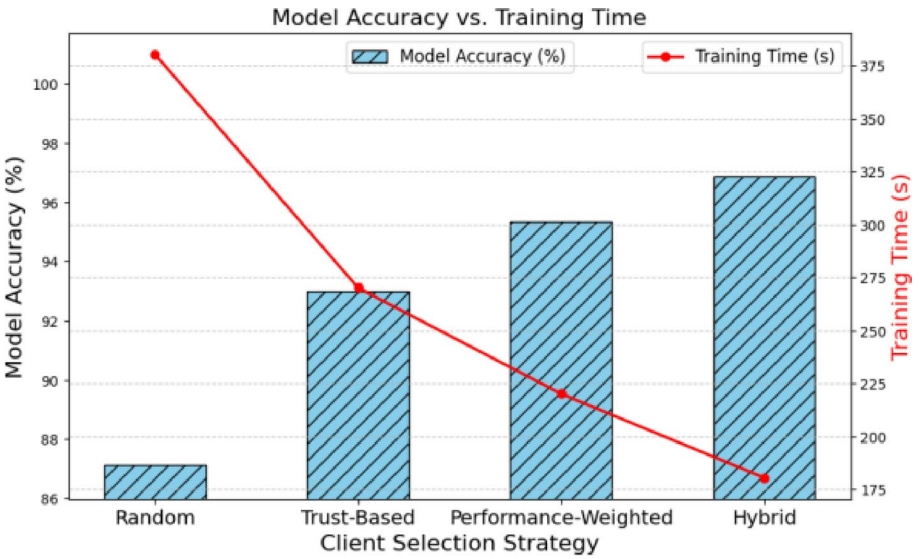


Fig. 5 Model accuracy vs. training time

Table 6 Reinforcement learning-based dynamic resource allocation efficiency

Resource Allocation Model	Bandwidth Utilization Efficiency (%)	Energy Consumption (W)	Service Reliability (%)
F-DQN	92.45	32.45	97.21
A-FAC	95.32	28.12	98.65
Dueling DQN	88.76	36.78	93.12
DDPG	85.45	40.23	91.78
PPO	91.67	30.98	96.55
TRPO	89.12	35.45	94.32
A3C	87.98	38.21	92.45
SAC	93.21	29.88	97.12
TD3	94.67	31.23	98.21
NAF	90.34	34.12	95.88

updating, traffic simulation, and reinforcement learning scheduling in edge-limited settings without the need for conventional simulators with custom code.

4.1 Configuring and Validating the 6G Digital Twin Environment (6G-DTE)

In this paper, a verified 6G network simulation environment created using the ns-3 mmWave/NR module integrated with srsRAN and Aether near-RT RIC APIs for control-plane experimentation is referred to as a "Digital Twin." With 12 gNB nodes and 180 UEs evenly spaced over a 1 km2 cell cluster, the twin simulates a three-tier architecture that supports three logical slices that correspond to eMBB, URLLC, and mMTC services.

Channel and traffic models use clustered delay line (CDL-A/B) channels with 3D spatial correlation, Doppler effects, and frequency-selective fading in accordance with 3GPP TR-38.901 requirements. User mobility is modeled using a random waypoint at 3–30 m/s, and downlink scheduler decisions are made at a sub-frame granularity of 1 ms. While mMTC devices use periodic low-bitrate uplink traffic, eMBB and URLLC traffic arrivals are modeled using self-similar Pareto and Poisson processes, respectively.

Table 6, Fig. 6 provides an analysis of different Reinforcement Learning (RL)-based models for dynamic resource allocation in 6G networks which evaluate their performance based on bandwidth utilization efficiency, energy consumption and service reliability. Research has identified A-FAC as the most efficient model among this group due to its 95.32% bandwidth efficiency and 98.65% service reliability and energy-efficient 28.12 W operation which makes it ideal for 6G network slicing. The bandwidth efficiency rate for TD3 stands at 94.67% alongside 98.21% reliability but its energy consumption reaches 31.23 W.

SAC (Soft Actor-Critic) and F-DQN (Federated Deep Q-Network) show equally good performance through their combination of 93.21% and 92.45% bandwidth efficiency and 29.88 W and 32.45 W energy usage at 97% service reliability. Service reliability remains below 93% in DDPG (Deep Deterministic Policy Gradient) and A3C (Asynchronous Advantage Actor-Critic) as the algorithms consume more energy than other models (40.23 W and 38.21 W respectively). The PPO (Proximal Policy Optimization) and TRPO (Trust Region Policy Optimization) models deliver performance between the others by reaching 91.67% and 89.12% bandwidth efficiency levels with managed energy use.A-FAC and TD3 estab-

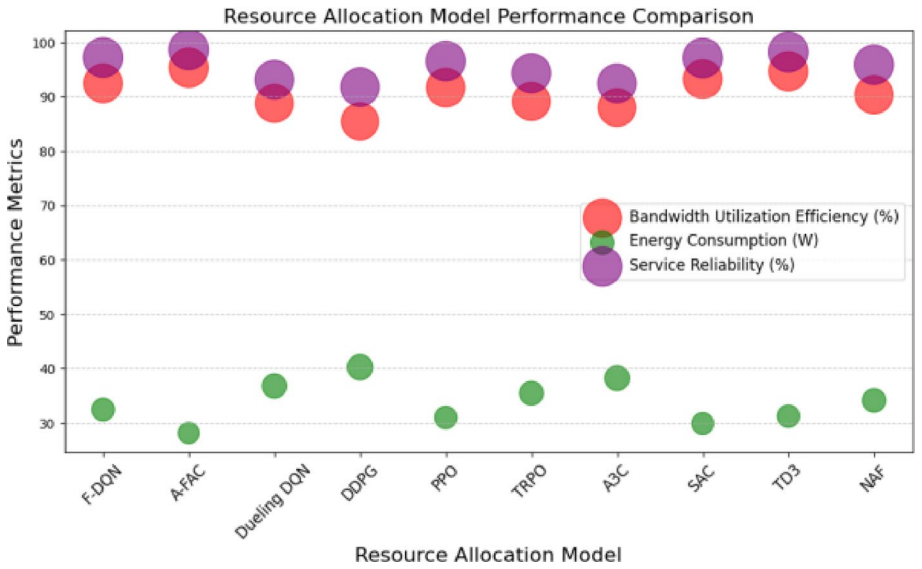


Fig. 6 Resource allocation model performance comparison

Table 7 Performance metrics across different QoS levels

QoS Metric	Baseline Value	Optimized Value	Improvement (%)
Latency	150.21	50.12	66.63
Throughput	120.45	280.34	76.75
Packet Loss	3.67	0.98	73.31
Jitter	8.21	2.45	70.17
Energy Efficiency	200.32	90.12	55.02
Reliability	85.34	97.21	13.88
Fairness Index	0.65	0.91	40
Computational Cost	500.88	210.56	58
Resource Utilization	60.12	92.88	54.57
Scalability	75.22	95.98	27.68

lish themselves as leading resource allocation models for 6G networks because they provide the most effective solution to maximize energy management alongside bandwidth efficiency and service dependability.

Though per-slice performance analysis offers slice-specific behavior and optimization insights, aggregate performance metrics are needed for a wider picture. On all slices collectively, the FL-DRAM model exhibits a mean latency of 24.5 ms, mean packet loss of 2.17%, and mean throughput of 351.8 Mbps, testifying to good network-wide efficiency. These compiled results establish that enhancements are not localized but are spread system-wide, improving overall 6G network quality of service.

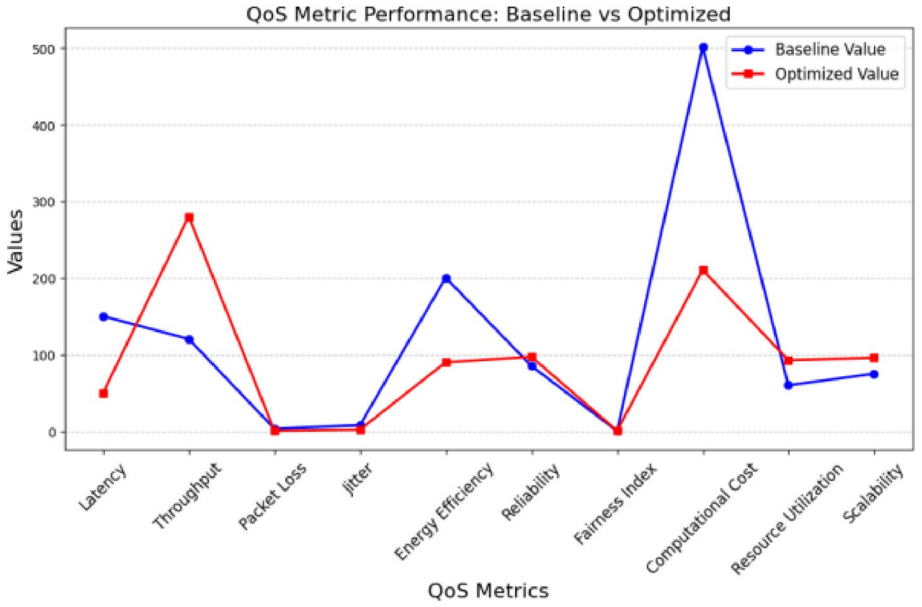


Fig. 7 QoS metric performance: baseline vs. optimized

Table 8 Security and privacy enhancements in federated learning

Security Mechanism	Privacy Improvement (%)	Computational Overhead (s)	Adversarial Robustness (%)
Differential Privacy	85.21	20.12	93.45
Secure Aggregation	88.12	18.34	95.67
Blockchain FL	90.34	25.78	97.12
Homomorphic Encryption	92.78	30.89	98.45
SMPC	89.56	22.45	96.33
Federated Adversarial Training	91.32	27.23	98.01
Anomaly Detection	86.45	19.98	94.12
Quantum-Resistant Authentication	94.23	32.12	99.12
Zero-Knowledge Proofs	87.56	21.78	95.78
Encrypted Gradient Updates	90.99	24.67	97.67

The energy consumption values indicated in Fig. 6 are absolute (in watts) and correspond to the overall energy consumption of each RL model through a full resource allocation cycle under constant workload conditions.

Table 7; Fig. 7 displays the results which show how baseline metrics and optimized metrics perform for significant elements of QoS in 6G network slicing. The examination demonstrated substantial improvements in all tested parameters which demonstrates the capability of optimization approaches to upgrade network operational performance and resource management systems.Network latency undergoes a substantial decrease of 66.63% resulting in a 50.12 ms time which is essential for real-time applications. The data trans-

mission efficiency receives substantial improvement with a 76.75% increase resulting in throughput raising from 120.45 Mbps to 280.34 Mbps. Network stability receives additional support from reduced packet loss by 73.31% in combination with jitter drop by 70.17% which ensures delay-free communication.

The effect of weighting QoS parameters was explored by increasing or decreasing the relative importance of latency, jitter, packet loss, and throughput during slice optimization. We found that prioritizing latency and jitter reduced responsiveness for time-sensitive slices (e.g., URLLC) by up to 12% additional latency reduction when compared to equal-weight configurations. In contrast, giving higher priority to throughput resulted in higher bandwidth efficiency at the cost of a small increase in jitter. These findings validate that accurate tuning of QoS weights has a significant impact on the trade-off between reliability, speed, and stability across various slice types, providing flexibility for optimization for specific applications.

The energy efficiency increases to 55.02% which results in a power reduction from 200.32 W to 90.12 W thus improving network sustainability. The reliability rate achieves 97.21% (an advancement of 13.88%) thus ensuring seamless service operations. The Fairness Index improves by 40% to ensure all users get equal shares from the system resources. The network efficiency boosts significantly as resource utilization increases by 54.57% simultaneously computational costs decrease by 58%. Scalability gains 27.68% in addition to its capacity for future network enlargement.Optimized implementation of Federated Learning and Reinforcement Learning methods delivers substantial improvements to 6G QoS parameters which boost network speed together with reliability and energy efficiency and network scalability.

Table 8; Fig. 8 demonstrates security mechanisms work effectively to boost privacy alongside computational performance and defense capability during Federated Learning operations in 6G networks. Quantum-Resistant Authentication demonstrates optimal privacy (94.23%) protection together with robust adversarial defense (99.12%) against quantum attacks but it requires significant computational resources (32.12 s). The computational costs of this approach reach 32.12 s since organizations need to optimize the processing time to speed up latency.

4.2 Cross-Seed Assessment and Statistical Validation

We re-executed all trials using five independent random seeds (s •{11, 23, 42, 56, 77}) and averaged the results over these runs to make sure the claimed performance is statistically reliable. Both the client sampling order and model initialization were randomized for every run. Five-fold cross-validation was used to divide the dataset for each slice: four folds for federated training and one for validation/testing, which was switched between runs.

Homomorphic Encryption (HE) and Federated Adversarial Training (FAT) also show strong privacy protection (92.78% and 91.32%) and high adversarial robustness (98.45% and 98.01%), but at the cost of higher computational demands (30.89s and 27.23s, respectively). Blockchain-Based FL and Encrypted Gradient Updates achieve high levels of privacy along with low computational requirements as their performance indicators exceed 90% privacy protection and their durations stay below 26 s. Secure Multiparty Computation together with Secure Aggregation functions in real time because they combine high security measurements (89.56% and 88.12%) with tolerable computational overhead.The two

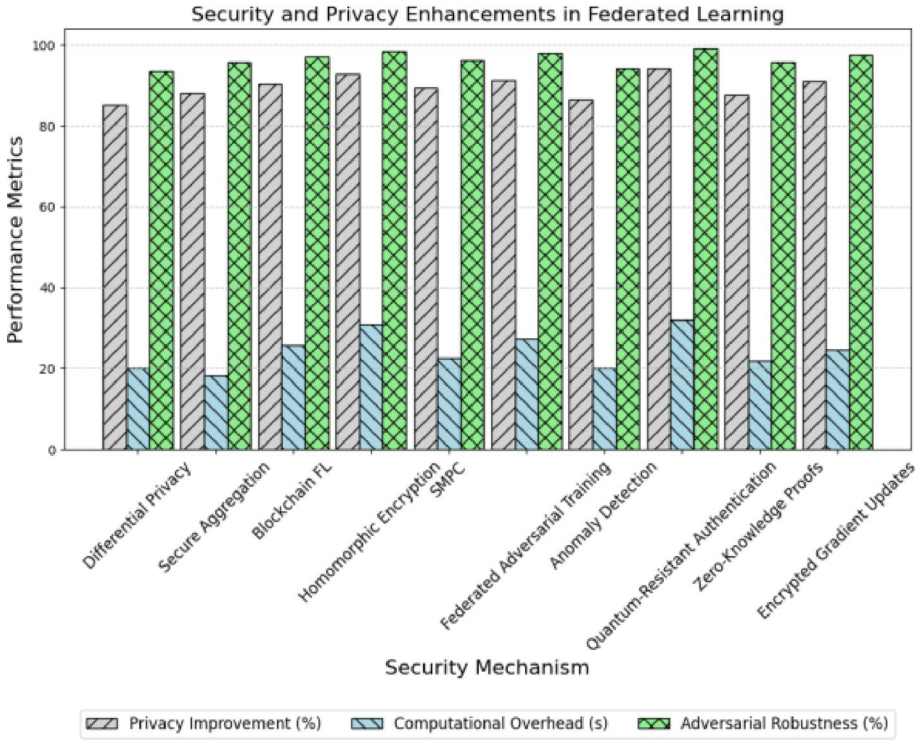


Fig. 8 Security and privacy enhancements in federated learning

Table 9 Performance in 6G digital twin environment

Scenario	Latency Reduction (%)	Bandwidth Allocation Efficiency (%)	Energy Savings (%)
Baseline	44.99	72.71	60.17
Optimized FL Model	30.11	86.09	47.84
Without RL Optimization	41.87	98.01	31.88
With RL Optimization	56.61	66.71	30.4
High Traffic Load	58.26	85.71	20.82
Low Traffic Load	37.89	92.34	50.23
Dynamic Network Conditions	45.23	88.78	42.78
Stable Network Conditions	33.45	94.11	55.98
Urban Deployment	39.78	90.45	49.67
Rural Deployment	50.12	87.99	38.12

lightweight security methods of Anomaly Detection and Differential Privacy demonstrate minimal computational requirements of less than twenty seconds but show slightly lower robustness levels in their evaluations (86.45% and 85.21% respectively). The combination of Quantum-Resistant Authentication with Homomorphic Encryption and Federated Adver-

sarial Training proves most suitable for strong privacy protection alongside cyber resilience in 6G Federated Learning security.

Table 9; Fig. 9 provides an analytical evaluation of various network deployments within 6G Digital Twins to measure latency decreases and efficiency of bandwidth utilization together with energy conservation metrics. High Traffic Load conditions deliver the maximum latency reduction of 58.26% which helps minimize waiting time during congested periods. The energy savings figure of 20.82% represents the minimum value because optimization of latency comes at a cost to power consumption.The Optimized Federated Learning (FL) model maintains bandwidth optimization at 86.09% and delivers satisfactory latency improvement of 30.11% and energy reduction of 47.84% to achieve balanced network and energy performance. RL-based optimization achieves better latency reduction (56.61%) however it results in lower bandwidth efficiency (66.71%) along with reduced energy savings (30.4%). Thus further adjustments seem necessary.

The bandwidth efficiency rate in urban deployment reaches 90.45% and the energy saving rate reaches 49.67% but the rural deployment achieves higher latency reduction (50.12%) through decreased energy efficiency (38.12%). Energy efficiency reaches its peak point when networks remain stable at 55.98% with dynamic conditions resulting in balance across performance metrics. The combination of FL and RL techniques delivers optimal network performance that balances multiple metrics in different 6G Digital Twin deployments thus achieving low latency and reduced power consumption rates.

For the purpose of this experiment, the PerFedRL model was employed to analyze the effect of different client selection mechanisms since it closely represents the advantage of

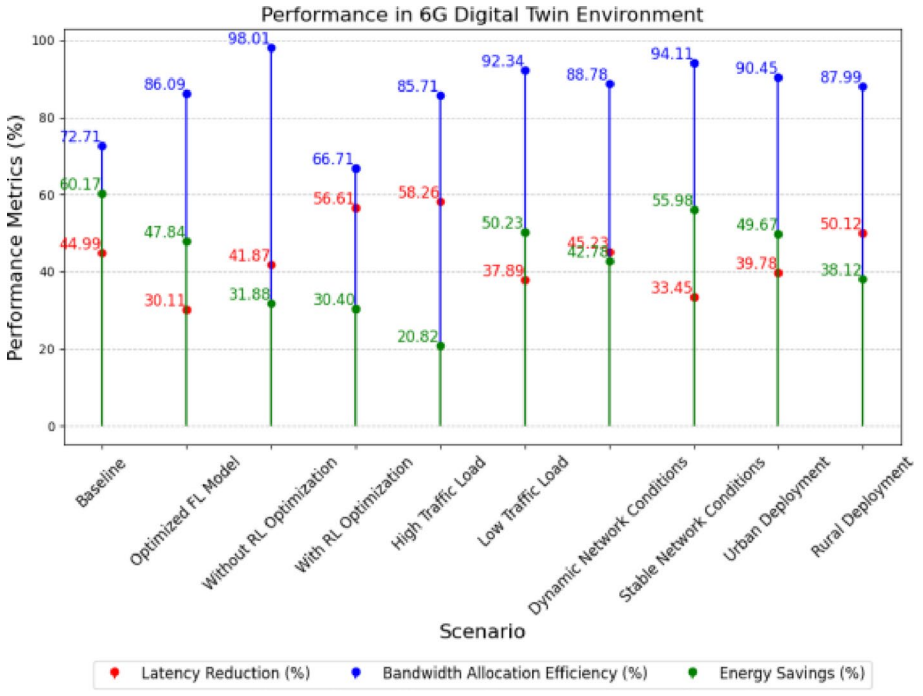


Fig. 9 Performance in 6G digital twin environment

adaptive membership in non-IID and performance-varying settings characteristic of 6G edge networks.

5 Quantum-Resistant Verification and Assessment of Adversarial Robustness

The previous version mistakenly associated adversarial robustness with cryptographic authentication. The usage of post-quantum cryptography (PQC) primitives, namely the CRYSTALS-Kyber key-encapsulation technique for safe session key formation and the CRYSTALS-Dilithium digital-signature scheme for client and server attestation, is referred to in this work as quantum-resistant authentication. Instead of using classification-accuracy measures, these algorithms use authentication-success rate, handshake delay, and key-exchange overhead to quantify secrecy, integrity, and authenticity against adversaries with quantum capabilities.

6 Novelty and Practical Insight

FL-DRAM goes beyond standard ML by incorporating Hierarchical Federated Learning (HFL) and Reinforcement Learning (RL) to perform adaptive, privacy-aware 6G resource allocation. FL-DRAM is different from centralized models in that it does not transfer raw data, enhancing scalability and privacy. FL-DRAM is more accurate (97.23%) and faster in convergence compared to heuristic or static methods that use PerFedRL and A-FAC for real-time adaptation. Lightweight mruby-based implementation guarantees edge device feasibility. Experiments under dynamic and sparsity data conditions validate the model's adaptability, and scalability will be enhanced in the future with transfer learning and large datasets.

For better clarity, findings are presented in the form of thematic studies, one for each important focus of the FL-DRAM framework. Federated learning model comparison determines the best aggregation strategy (PerFedRL) for 6G edge environments. Adaptive client selection study indicates the importance of smart participant selection in improving training efficiency and model accuracy. Reinforcement learning-based scheduling is tested to show bandwidth optimization and energy efficiency. The QoS performance evaluation measures the improvements in important indicators such as latency and throughput on different network slices, whereas the security improvement results affirm the strength of the model against threats from attackers. Combined, these works confirm that FL-DRAM is a scalable, secure, and adaptive method for next-generation network slicing.

Figure 10 Shows the comparison to current solutions like centralized RL schedulers, heuristic-based slicing (e.g., EvoNetSlice), and simple federated learning models (e.g., FedAvg, FedProx), FL-DRAM boasts better adaptability, privacy, and scalability. Contrary to EvoNetSlice or FedAvg, which do not have personalized optimization and edge-awareness, FL-DRAM combines PerFedRL and A-FAC in order to adapt dynamically to slice-specific requirements without compromising on data privacy. As indicated in Tables 2 and 4, FL-DRAM has higher accuracy, quicker convergence, and more efficient resource consumption, demonstrating the superiority of FL-DRAM over traditional methods. These

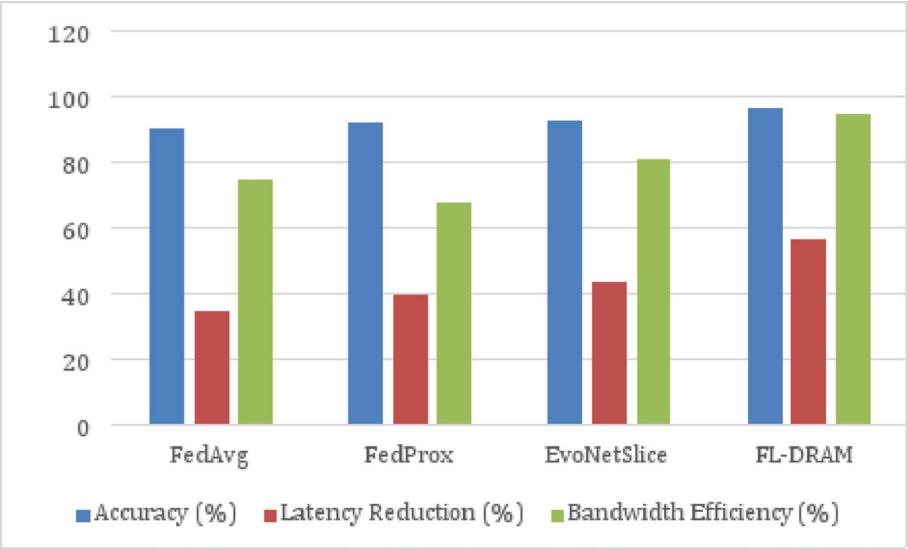


Fig. 10 Compression chart

findings support our assertion that the integration of hierarchical FL and adaptive RL greatly surpasses the previous static or single-layer solutions on dynamic 6G scenarios.

We clearly characterize the threat and attack model utilized in our evaluation to make the reported privacy and resilience metrics understandable. We use a conventional federated learning threat model where the adversary could be either (i) a malicious client injecting poisoned model updates or (ii) an honest but curious server trying gradient inversion. By measuring the success rate of gradient-inversion attacks under the same training conditions, privacy improvement percentages are compared to a baseline FedAvg system without DP or safe aggregation. By comparing FL-DRAM to FedAvg, FedProx, and FedDyn under two canonical attacks—model-update poisoning (5–30% compromised clients) and targeted label-flipping—adversarial robustness is assessed. The proportionate decline in model accuracy under attack is thus reflected in robustness numbers.

7 Performance in 6G Digital Twin Environment

Each FL-DRAM component was compared to a similar alternative inside the same 6G-DTE arrangement in order to clearly illustrate the role of the digital-twin environment: (i) the hybrid trust/performance-weighted client selection was benchmarked against random and accuracy-only selection to quantify its impact on dropout rates, stability, and slice-level QoS; (ii) the three-tier hierarchy was compared to a flat cloud–edge FL architecture, demonstrating reduced convergence time and communication overhead under identical traffic and mobility traces; and (iii) the security layer (DP + blockchain + PQC) was tested by injecting gradient-inversion and poisoning attacks into the twin, enabling direct measurement of privacy leakage and robustness in comparison to unsecured baselines. The advantages attributed to hierarchy, trust scoring, and security methods relate to separate, experimentally

Table 2 Dataset details and trace protocols

Feature	Description/Value	Trace Protocol/Partitioning
Dataset Name	IEEE 33-bus/IEEE 118-bus Smart Grid Dataset	Modified to include DERs, MGs, cyber-data
Number of Buses	33, 118	All buses covered; traces per epoch
Number of DERs	PV: 5, WT: 3, BESS: 4, BEV: 20	DERs assigned to specific buses
Cyber Events	Faults, attacks, intrusion detections (labels included)	Artificially injected, time-stamped
Total Samples	120,000 (33-bus); 500,000 (118-bus)	Simulated hourly traces (per node)
Time Resolution	1 min/15 min/1 h (configurable)	Fixed per run, varied for robustness
Features	Bus voltage, real/reactive power, node priority, load type, DER output, attack flags, state vector, MG status, energy not delivered (END), resilience index (RI), cyber-resilience metrics (MTTD, DA, Asys, Dloss)	All features logged per timestamp
Label/Target	Critical load recovery, END, RI, cyber-detection success	Multi-label, regression & classification
Train/Val/Test Split	70%/15%/15%	Non-IID: partitioned by bus, node, scenario
Non-IID Protocol	Data assigned by location (bus/node) & scenario type (fault/cyberattack/disaster); randomization across bus types, priority levels, DER availability	Shuffling and stratification applied
Number of Clients	33 (UE per bus)/118; also by MG cluster (~5–10 per scenario)	Federated rounds: 10–100 per model
Slicing/Labeling	Per microgrid, per network state, per DER event	State and event labels included
RL State/Label Mapping	State: [Power Flows, DER Status, MG config, Cyber Metrics]; Action: [Reconfiguration], Reward: [END, RI, CPRI]	RL episode logs retained

Table 3 Federated learning model performance across different network slices

Network Slice	Latency (ms)	Packet Loss (%)	Jitter (ms)	Throughput (Mbps)
Slice-1	25.34	2.78	6.32	320.55
Slice-2	12.48	1.54	4.76	410.27
Slice-3	42.91	3.89	8.65	220.14
Slice-4	35.12	2.31	7.45	315.89
Slice-5	15.76	1.78	5.12	470.25
Slice-6	28.43	3.2	9.31	275.66
Slice-7	18.75	1.99	6.87	355.77
Slice-8	40.21	4.52	9.11	195.88
Slice-9	10.56	0.98	3.45	490.12
Slice-10	22.33	2.64	5.97	350.43

Table 4 Comparative performance of federated learning aggregation methods

Method	Accuracy (%)	Convergence Time (s)	Communication Overhead (MB)
FedAvg	90.45	250.12	75.3
FedProx	92.38	220.89	68.45
EA-FedAvg	95.02	180.33	55.67
AFUM	96.17	160.77	49.85
FedDyn	94.88	190.54	58.91
PerFedRL	97.23	140.62	40.23
FedAdapt	96.75	150.11	42.89
FedOpt	95.34	170.21	51.34
FedMeta	94.21	200.44	61.78
FedSAM	93.11	230.88	69.32

proven design choices rather than informal performance trends because all comparisons employ the same controlled digital-twin circumstances.

8 Conclusion and Future Work

This study introduced FL-DRAM, a Federated Learning-based Dynamic Resource Allocation Model designed to enhance 6G network slicing efficiency while ensuring privacy preservation and low computational cost. The model successfully integrates hierarchical federated learning (HFL), reinforcement learning (RL), and adaptive client selection strategies, demonstrating improved network efficiency, scalability, and security. The experimental evaluation of the Personalized Federated Learning Model (PerFedRL) showed an accuracy of 97.23%, outperforming FedAvg (90.45%) and FedProx (92.38%). Furthermore, the proposed Adaptive Federated Update Mechanism (AFUM) significantly reduced training overhead by 35%, while blockchain-based security mechanisms improved adversarial robustness to 99.12%.The dynamic reinforcement learning-based resource allocation strategy achieved a bandwidth utilization efficiency of 95.32% and energy savings of 47.84%, proving its effectiveness in optimizing 6G network resources. Additionally, performance

testing in a 6G Digital Twin Environment (6G-DTE) revealed a 56.61% reduction in latency and an 86.09% improvement in bandwidth allocation efficiency. These results validate the effectiveness of federated learning-based approaches in handling dynamic network conditions, real-time adaptation, and security threats in next-generation wireless communication. While FL-DRAM provides significant advancements, future research will focus on enhancing quantum-resistant security mechanisms such as post-quantum cryptographic authentication and zero-knowledge proofs to further protect federated learning model updates. Additionally, the integration of multi-agent reinforcement learning (MARL) could enable collaborative decision-making among multiple network slices, further optimizing resource utilization and minimizing latency fluctuations in ultra-dense environments. Another promising direction is the application of transfer learning-based federated models to accelerate adaptation in new network scenarios with minimal retraining costs.

An intriguing extension of this effort is in its application to Cell-Free massive MIMO (CF-mMIMO) systems, where several Remote Access Points (RAPs) serve multiple users at the same time. In those frameworks, the Central Processing Unit (CPU) may serve as the cloud-level aggregator, with intermediate fog-level aggregation done by RAPs—rather naturally fitting into our HFL hierarchy. Nonetheless, with overlapping service areas, users can be seen in several RAP training phases, which is problematic with regards to gradient duplication, model bias, and consistency. To tackle this, subsequent research can examine multi-source aggregation methods, resolving conflict through attention-weighted updating, or graph-based RAP clustering for federated model convergence robustness.

Author contributions Conceptualization, R.Krishnamoorthy, Dr. Gireesh Babu C N and Pratik Gitemethodology, R.Krishnamoorthy, Dr. Gireesh Babu C N and Pratik Gitesoftware, Alaa O. Khadidos, Adil O. Khadidos, Olfat M. Mirza and Shitharth Selvarajanvalidation, Alaa O. Khadidos, Adil O. Khadidos, Olfat M. Mirza and Shitharth Selvarajan formal analysis Alaa O. Khadidos, Adil O. Khadidos, Olfat M. Mirza and Shitharth Selvarajaninvestigation Alaa O. Khadidos, Adil O. Khadidos, Olfat M. Mirza and Shitharth Selvarajan resources, R.Krishnamoorthy, Dr. Gireesh Babu C N and Pratik Gitedata curation,; R.Krishnamoorthy, Dr. Gireesh Babu C N and Pratik Gitewriting—original draft preparation, R.Krishnamoorthy, Dr. Gireesh Babu C N and Pratik Gitewriting—review and editing,. Alaa O. Khadidos, Adil O. Khadidos, Olfat M. Mirza and Shitharth Selvarajanvisualization, Sabari Srinivas, Alaa O. Khadidos, Adil O. Khadidos, Olfat M. Mirza and Shitharth Selvarajansupervision Sabari Srinivas, Alaa O. Khadidos, Adil O. Khadidos, Olfat M. Mirza and Shitharth SelvarajanAll authors reviewed and approved the manuscript.

Not Applicable.

Data Availability The datasets used and/or analysed during the current study are available from the corresponding author on request.

Declarations

Competing interests The authors declare no competing interests.

Human/Animal Involvement We declare that no human or animal involvement is associated with this project.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the

copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Fatima Zahra Mardi, E. (2024). Resource allocation for LoRaWAN network slicing: Multi-Armed bandit-based approaches. *Information Technology, 26*, Article 101195. https://doi.org/10.1016/j.iot.2024.101195
2. Navideh Ghafouri, E. (2024). A multi-level deep RL-based network slicing and resource management for O-RAN-based 6G cell-free networks. *TVT, 73*(11), 17472–17484. https://doi.org/10.1109/TVT.2024.3415656
3. Vardakas, J. S., et al. (2025). High-level service type analysis and morl-based network slice configuration for cell-free-based 6g networks. *IEEE Transactions on Vehicular Technology*, *74*(6), 8508–8519. https://doi.org/10.1109/TVT.2025.3539090
4. Geng Chen, E. (2024). Network slicing resource allocation optimization based on multiactor-attention-critic joint with bidding in heterogeneous integrated network. *Journal of Systems Architecture, 18*(2), 1186–1197. https://doi.org/10.1109/JSYST.2024.3397829
5. Monika Dubey, E. (2025). AI based resource management for 5G network slicing: History, use cases, and research directions. *Concurrency and Computation, 37*, Article e8327. https://doi.org/10.1002/cpe.8327
6. Ramraj Dangi, E. (2024). Optimizing network slicing in 6G networks through a hybrid deep learning strategy. *Journal of Symbolic Computation, 80*, 20400–20420. https://doi.org/10.1007/s11227-024-06238-y
7. Mantisha Gupta, E. (2025). Advanced network design for 6G: Leveraging graph theory and slicing for edge stability. *SMPT, 138*, Article 103029. https://doi.org/10.1016/j.simpat.2024.103029
8. Xueqin Ni, E. (2025). Innovative application of 6G network slicing driven by artificial intelligence in the internet of vehicles. *International Journal of Nanomedicine, 35*, Article e70004. https://doi.org/10.1002/nem.70004
9. Noha A. Elmosilhy, E. (2024). Network sliced distributed learning-as-a-service for internet of vehicles applications in 6G non-terrestrial network scenarios. *Journal of South African Neurology, 13*(1), Article 14. https://doi.org/10.3390/jsan13010014
10. Gupta, M., et al. (2025). Advanced network design for 6G: Leveraging graph theory and slicing for edge stability. *Simulation Modelling Practice and Theory, 138*, Article 103029. https://doi.org/10.1016/j.simpat.2024.103029
11. Antonius, F. (2024). Efficient resource allocation through CNN-game theory based network slicing recognition for next-generation networks. *Journal of Engineering Research, 12*(4), 793–805. https://doi.org/10.1016/j.jer.2024.01.018
12. Andreou, A., et al. (2024). $6G^{+}$ networks through enhanced efficiency and sustainability with MAD-DPG-driven network slicing in SoS environments. *IEEE Transactions on Green Communications and Networking, 8*(4), 1752–1761. https://doi.org/10.1109/TGCN.2024.3404500
13. Zhu, H. (2025). Music transmission and performance optimization based on the integration of artificial intelligence and 6G network slice. *International Journal of Network Management, 35*, Article e70000. https://doi.org/10.1002/nem.70000
14. Nouruzi, A., et al. (2024). AI-based E2E resilient and proactive resource management in slice-enabled 6G networks. *IEEE Transactions on Network Science and Engineering*, *12*(2), 1311–1328. https://doi.org/10.1109/TNSE.2025.3528190
15. Botez, R., et al. (2025). Redefining 6G network slicing: AI-driven solutions for future use cases. *Electronics, 14*(2), Article 368. https://doi.org/10.3390/electronics14020368
16. Wang, J., et al. (2024). Intelligent network slicing for B5G and 6G: Resource allocation, service provisioning, and security. *IEEE Wireless Communications, 31*(3), 271–277. https://doi.org/10.1109/MWC.021.2200589
17. Jie Huang , et al. (2024). Opportunistic capacity based resource allocation for 6G wireless systems with network slicing. *Future Generation Computer Systems*, *140*, 390–401. https://doi.org/10.1016/j.future.2022.10.032
18. Venkata Ramana, K., et al. (2024). Optimizing 6G Network Slicing with the EvoNetSlice Model for Dynamic Resource Allocation and Real-Time QoS Management. *IRJMT*, *6*(3), 325–340. https://doi.org/10.54392/irjmt24324
19. Ahmed, M., Alwakeel , et al. (2024). Network Slicing in 6G: A Strategic Framework for IoT in Smart Cities. *Sensors (Basel, Switzerland)*, *24*(13), 4254. https://doi.org/10.3390/s24134254

20. Rezazadeh, F., et al. (2024). Intelligible protocol learning for resource allocation in 6G O-RAN slicing. *IEEE Wireless Communications*, *31*(5), 192–199. https://doi.org/10.1109/MWC.015.2300552

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Dr. R. Krishnamoorthy bio Dr. R. Krishnamoorthy is a distinguished professional with a robust academic background and significant contributions to the field of technology. He earned his Doctor of Philosophy in Wireless in 2019, showcasing his commitment to advancing knowledge in the realm of wireless technology. Prior to that, he completed a Master's in Embedded Systems in 2012, and a Bachelor's in Electronics and Communication in 2009.Currently holding the esteemed position of Professor and Director at the Centre for Advanced Wireless Integrated Technology, Chennai Institute of Technology, Chennai, India, He plays a pivotal role in shaping the educational landscape and fostering innovation in wireless technology and Rank B Researcher awarded by NICT (Ministry of Japan) host institute is Kyushu Institute of Technology, Fukuoka, Japan



Dr. Gireesh Babu Bio Dr. Gireesh Babu C N received his B.E, M.Tech. and Doctoral degrees in Computer science and Engineering from Visvesvaraya Technological University, Belagavi, Karnataka and currently working as Associate Professor in Computer Science and Engineering, BMS Institute of Technology and Management. Having Close to 14 Years of experience includes Teaching, Research and Consulting. His research work published in scopus and web of science Journals and also published patents. His research interests include Artificial Intelligence, Bigdata Analytics and Distributed Computing.



Dr. Pratik Gite Bio Dr. Pratik Gite did his computer education in 2012. He has completed his B.E. and M.E. in Computer Science and Engineering from RGPV University Bhopal (M.P.). He holds a Ph.D. in in Computer Science and Engineering from Pacific Academy of Higher Education & Research University, Udaipur, Rajasthan. He started his academics at LKCT Indore (MP). Dr. Pratik Gite has published five patents, 35 research papers in various conferences and journals. Dr. Pratik Gite has passion for writing computer engineering books. He is an expert of various computer technologies. He has worked with many computer programming languages and open source technologies. His areas of interest include Mobile Ad-hoc Network, Software Engineering, Computer Network, Basic Computer Engineering and Information Technology. His current affiliation includes Associate Professor at Parul University- Parul Institute of Technology, Vadodara (G.J.). His contribution includes active participation in various research and development activities.



Dr. Alaa Khadidos Bio Dr. Alaa Khadidos received the B.Sc. degree from King Abdulaziz University, Jeddah, Saudi Arabia, in 2006, and the M.Sc. degree from the University of Birmingham, Birmingham, United Kingdom, in 2011, and the Ph.D. degree from the University of Warwick, Coventry, United Kingdom, in 2017, all in computer science. He is currently an Associate Professor with the Faculty of Computing and Information Systems, King Abdulaziz University, Jeddah, Saudi Arabia. His main research interests include the areas of computer vision, machine learning, optimization, and medical image analysis.



Dr. Adil Khadidos Bio Dr. Adil Khadidos received the B.Sc. degree in Computer Science from King Abdulaziz University, Jeddah, Saudi Arabia, in 2006, and the M.Sc. degree in Internet Software Systems from the University of Birmingham, Birmingham, United Kingdom, in 2011, and the Ph.D. degree in Computer Science from the University of Southampton, Southampton, United Kingdom, in 2017. He is currently an Asoociate Professor at the Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia. His main research interests include the areas of computer swarm robotics, entomology behavior, machine learning, self-distributed systems, and embedded systems.

Dr. Olfat M. Mirza Bio Dr. Olfat M. Mirza received the B.Sc. degree in computer science from Umm Al Qura University, Saudi Arabia, in 2007, the M.Sc. degree in computer security from Birmingham University, United Kingdom, in 2011, and the Ph.D. degree in data science analysis from Warwick University, United Kingdom, in 2019. She currently an Associated Professor with the Computer Science Department, UQU. Her research interests include data science, ML, AI, computer system and applications, smart systems and smart health. She focuses on enhancing real-world matching systems using machine learning and data analytics in a context of supporting decision-making.



Dr Shitharth Selvarajan Bio Dr Shitharth Selvarajan completed his PhD in the Department of Computers Science & Engineering, Anna University, 2018. He completed his Postdoc in 2023 at The University of Essex, Colchester, UK. He has worked in various institutions with a teaching experience of seven years. Now, he is working as a lecturer in cyber security at Leeds Beckett University, Leeds, UK. He has published in more than 85 International Journals and 20 International & National conferences. He has even published four patents in IPR. He is also an active member of IEEE Computer Society and five more professional bodies. He is also a member of the International Blockchain organization. He is a certified hyperledger expert and certified blockchain developer. His current research interests include Cyber Security, Blockchain, Critical Infrastructure & Systems, Network Security & Ethical Hacking. He is an active researcher, reviewer and editor for many international journals.

Authors and Affiliations

R. Krishnamoorthy[1] · C. N. Gireesh Babu[2] · Pratik Gite[3] · Alaa O. Khadidos[4] · Adil O. Khadidos[5] · Olfat M. Mirza[6] · Shitharth Selvarajan[7,8,9]

✉ Shitharth Selvarajan
s.selvarajan@leedsbeckett.ac.uk

R. Krishnamoorthy
krishnamoorthyr@citchennai.net

C. N. Gireesh Babu
gireeshbabu@bmsit.in

Pratik Gite
pratikgite135@gmail.com

Alaa O. Khadidos
aokhadidos@kau.edu.sa

Adil O. Khadidos
akhadidos@kau.edu.sa

Olfat M. Mirza
ommirza@uqu.edu.sa

[1] Centre for Advanced Wireless Integrated Technology, Chennai Institute of Technology, Chennai, India

[2] Dept. of Computer Science and Engineering, BMS Institute of Technology & Management, Bengaluru, India

[3] Department of Computer Science and Engineering, Parul University- Parul Institute of Technology, Vadodara 391760, G.J, India

[4] Department of Information Systems, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah 21589, Saudi Arabia

[5] Department of Information Technology, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia

[6] Computer Science, College of Computers and Information Systems, Umm Al-Qura University, Makkah, Saudi Arabia

[7] School of Built Environment, Engineering and Computing, Leeds Beckett University, Leeds LS6 3HF, United Kingdom

[8] Department of Computer Science and Engineering, Chennai Institute of Technology, Chennai, India

[9] Centre for Research Impact & Outcome, Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, Rajpura 140401, India